

Module

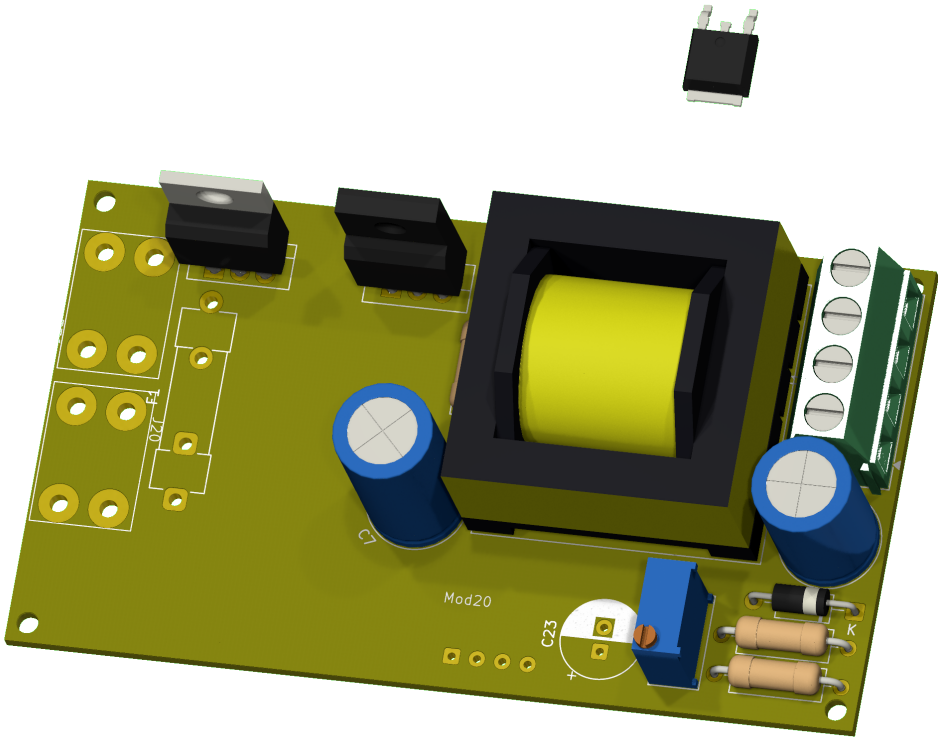
Variant: PRELIMINARY

Rev 1.0.0+ (Unreleased)

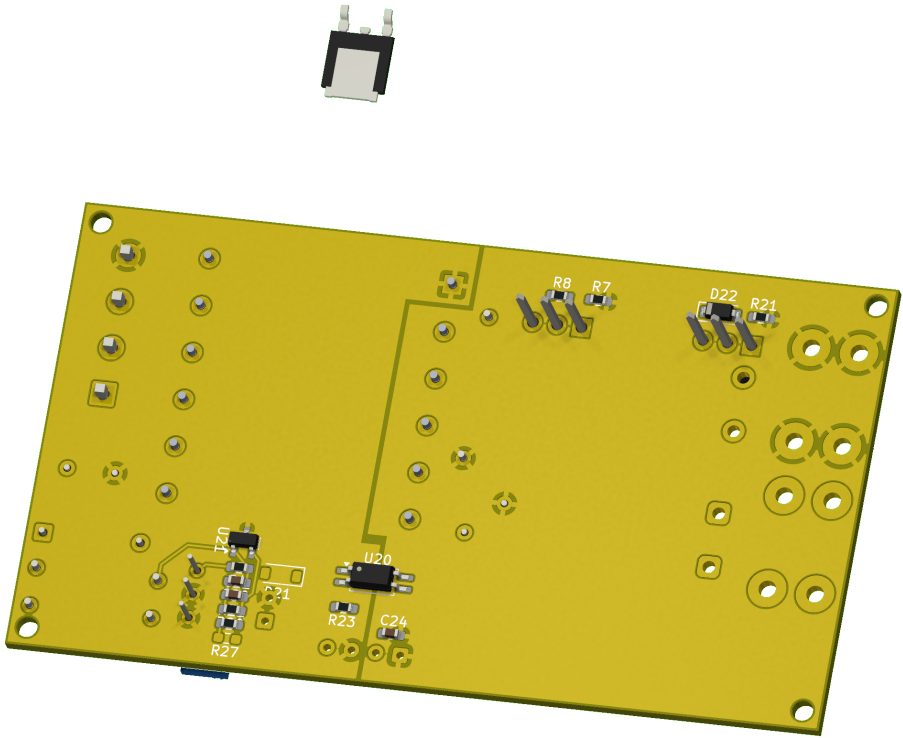
2026-01-28

Page	Index
1	Cover Page
2	Block Diagram
3	Schematic
4	Revision History
5	
6	
7	
8	
9	
10	

TOP VIEW



BOTTOM VIEW



DRAFT - Early concept stage, schematic drafted.

PRELIMINARY - Minor schematic changes are expected, ready for layout planning.

CHECKED - DRC completed, schematic and layout ready for review.

RELEASED - Design ready for manufacturing.

NOTE STYLING CONVENTION

DESIGN NOTE:
Example text for
informational
design notes.

DESIGN NOTE:
Example text for
debug notes.

DESIGN NOTE:
Example text for
cautionary
design notes.

DESIGN NOTE:
Example text for
critical design
notes.

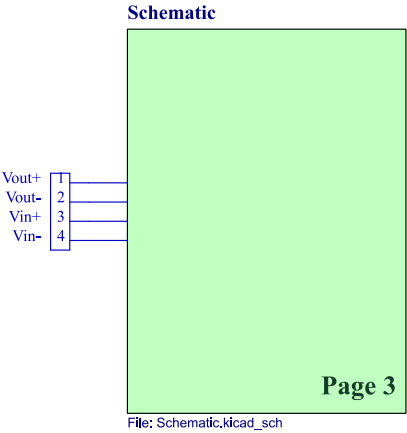
LAYOUT NOTE:
Example text for
critical layout
guidelines.

Date: 28-Jan-2026

	Comments:	Company: FR	Variant: PRELIMINARY	Git Hash: 120601b
	Sheet Title:	Board Name: Module	Project Name: Isolated step-up 12:450 V	
	Sheet Path: /	File Name: Isolated_12to450V.kicad_sch	Designer: FR	Date: 2025-12-20
			Reviewer: NA	Revision: 1.0.0+ (Unreleased)
			Size: A3	Sheet: 1 of 4

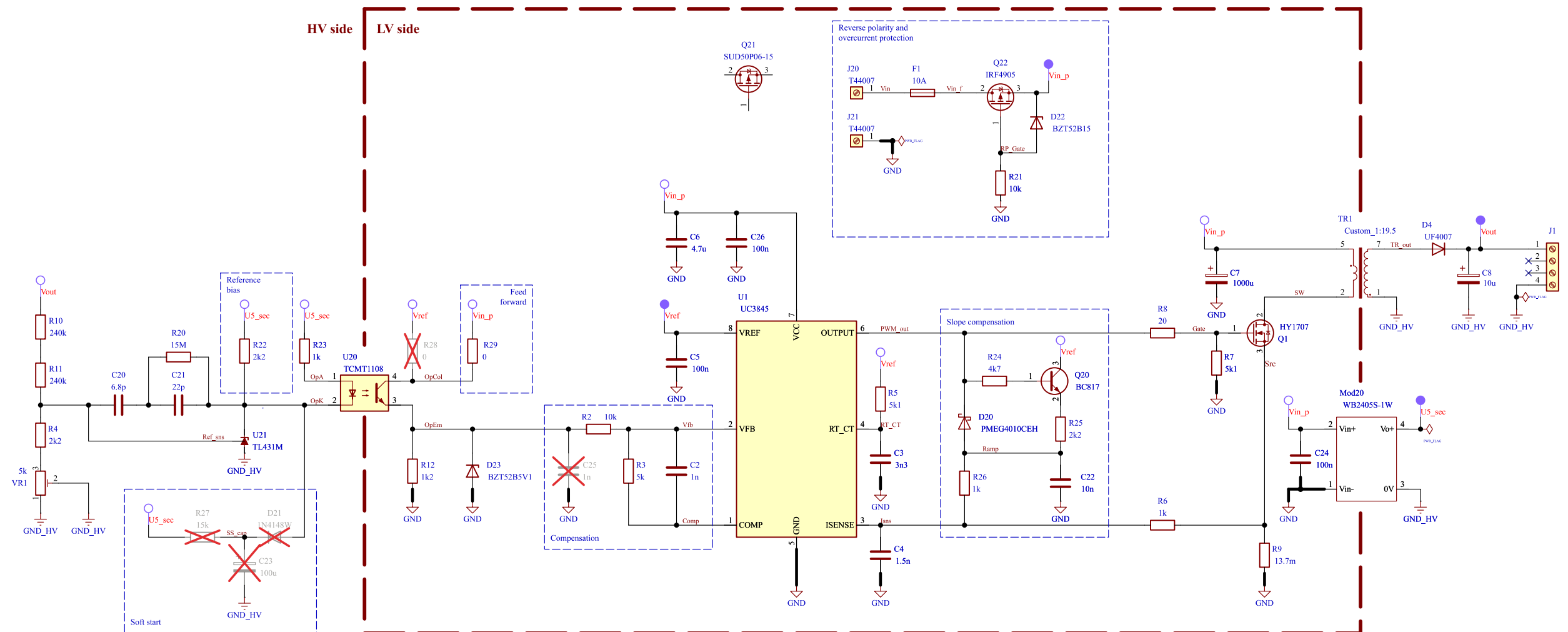
[2] Block Diagram

MODULE INTERFACE:
Vin: 9 V to 24 V according
to AliExpress product page
Vout: 200 V to 450 V,
adjustable through VR1



	Comments: AliExpress module example (link)		Company: FR		Variant: PRELIMINARY		Git Hash: 120601b			
			Board Name: Module			Project Name: Isolated step-up 12:450 V				
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch		Designer: FR		Date: 2025-12-20		Revision: 1.0.0+ (Unreleased)	
	Sheet Path: /Block Diagram/				Reviewer: NA		Size: A4		Sheet: 2 of 4	

[3] Schematic



U21 uses TL432 pinout - Verify if pinning matches in case that part is replaced.

R4 limits the max. output voltage to approximately 545 V
VR1 allows adjusting the output voltage down to approximately 169 V

- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Estimated specs for unidentified / custom parts:
- R9: ~13.7 mΩ (110 mV @ 8.04 A)
- TR1: ~1:19.5 turns ratio, 3.54:1453 uH

Sheet Title: Schematic	Comments:		Company: FR		Variant: PRELIMINARY	Git Hash: 120601b
	Sheet Path: /Block Diagram/Schematic/		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	File Name: Schematic.kicad_sch		Designer: FR	Date: 2025-12-20	Revision: 1.0.0+ (Unreleased)	
Reviewer: NA		Size: A3	Sheet: 3 of 4			

[4] Revision History

Version 1.0.0 - 2026-01-24

Cloned non-isolated design revision 1.1.0 as a working base.

Updated R10 and R11 to 240 kΩ (cloned design incorrectly used 140 kΩ).

Updated U1 to UC3845 (cloned design incorrectly used UC3843).

Reworked feedback loop to include opto-coupler while keeping the same reference on both sides to facilitate testing / reuse of original board.

Added two options of slope compensation / ramp injection (from U1 RT_CT pin and from U1 OUTPUT pin).

Adjusted R2 to 10 kΩ, R3 to 5 kΩ and C2 to 1 nF to improve behaviour (output swaps from bursts to 0 % duty periodically).

Added soft-start (D21, R27 and C23).

Increased C5 to 100 nF.

Adjusted C4 to 1.5 nF and R6 to 1 kΩ (similar time constant) to facilitate ramp injection.

Updated parts placement where suitable to reuse existing layout and added an auxiliary perf-board to accomodate the new parts.

Version 1.0.1 - 2025-12-21

	Comments:	Company:		Variant:	Git Hash:
		FR		PRELIMINARY	120601b
		Board Name:		Project Name:	
		Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Revision History	Revision History.kicad_sch	FR	2025-12-20	1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
	/Revision History/		NA	A4	4 of 4